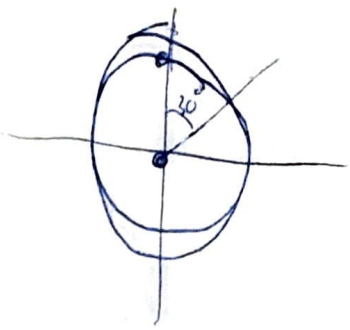
Nitya

W2-4

$e = 0$

Q24) $v = 7.5$ km/s

$\Delta i = 30^\circ$ - single impulse



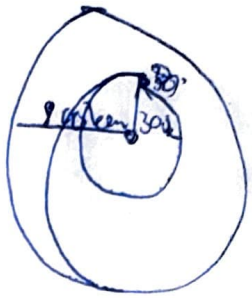
$a_{before} = a_{after}$

$\Delta V = ?$ $\quad \Delta V_s = 2V_i \sin\left(\frac{\theta}{2}\right)$

$= 2 \times 7.5 \times \sin(15^\circ) = \boxed{3.8822 \text{ km/s}}$

Q25) Hohmann: $\mu_\oplus = 398600 \text{ km}^3/\text{s}^2$ $\quad R_\oplus = 6378$

W2-L1



300 km $\quad$ 800 km

$R_1 = 6678$ km $\quad R_2 = 7178$ km

$a_t = \frac{R_1 + R_2}{2} = \frac{13856}{2} = 6928$ km

$V_1 = \sqrt{\frac{\mu}{R_1}} = \sqrt{\frac{398600}{6678}} = 7.7258$ km/s

$\varepsilon_t = -\frac{\mu}{2a_t} = -\frac{398600}{2 \times 6928} = -28.7673$

$V_{t_1} = \sqrt{2\left(\frac{\mu}{R_1} + \varepsilon_t\right)} = \sqrt{2(59.688528)} = 7.86395$ km/s

$\Delta V_1 = |7.86395 - 7.7258| = 0.13819$ km/s

$V_2 = \sqrt{\frac{\mu}{R_2}} = 7.451898$ km/s

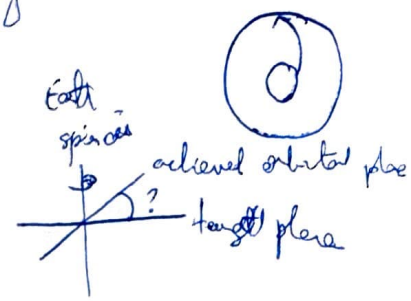
$V_{t_2} = \sqrt{2\left(\frac{\mu}{R_2} + \varepsilon_t\right)} = \sqrt{2(55.536128.7673)} = 7.3162$ km/s

$\Delta V_2 = 0.1356847$ km/s

$\Delta V = \Delta V_1 + \Delta V_2 = 0.27387$ km/s

$= \boxed{273.87 \text{ m/s}}$

Q26) Liftoff = +6 mins after launch site was carried through the target plane by Earth's rotation $= 360s$



RAAN error = ?

* use sideral rotation rate

(given) sideral day = 23h 56m 4s = 86,164.09s

Sidereal rotation rate = $\omega_{\oplus} = 7.2921 \times 10^{-5}$ rad/s

$= 0.004178$ °/s

$\ddot{x} - 3n^2x - 2n\dot{y} = 0$

$\ddot{y} + 2n\dot{x} = 0$

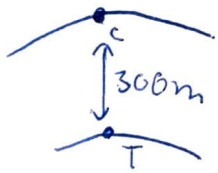
$\dot{y} = -3nx_0/2$

Launch delay = 360s

$\Delta\Omega = \omega_{\oplus} \times \Delta t = 0.004178 \times 360 =$ **1.5041°**

Q27)

W4-L1



$T_{target} = 95$ mins $= 5700$

2 target orbit $\rightarrow T_{ch} = 11,400s$

$\dot{x}_0 = 0$

$\dot{y}_0 = -\frac{3}{2}nx_0$

$a_1 = y_0$

$a_2 = 2x_0 - \frac{2\dot{y}_0}{n} = 0$

$a_3 = x - \frac{3}{2}\dot{y}_0 t + 3x_0 = 0$

$2a_0$

1 orbit period $\rightarrow 3\pi x_0$ move

2 h $\rightarrow 6 \times \pi \times 300$

$= 5654.866$ m

Nityan

Q28) chaser: $x_0 = +250\,m$

$y_0 = 0$

$\dot{x} = 0$

$\dot{y} = ?$ for bonded relative motion under HCW



n → mean motion

$n = \frac{2\pi}{T}$

bounded motion condition ← $\dot{y} = -2nx_0$

$n = \frac{2\pi}{T} \rightarrow \sqrt{\frac{\mu}{a^3}}$ $\quad T_{700km} = 2\pi\sqrt{\frac{a^3}{\mu}} = 2\pi\sqrt{\frac{(7078)^3}{398600}}$ ~~= 5926 secs~~

$\Rightarrow \dot{y} = -2x$

$n = \sqrt{\frac{398600\ km^3/s^2}{(7078)^3\ km^3}} = 0.0010602\ rad/s$

$\dot{y} = -2 \times 0.0010602 \times 250 = \boxed{-0.5301\ m/s}$

Q29) Chief: $a = 7000\,km$ $\quad e = 0.0015$ $\quad \omega = 0^\circ$

Deputy: $a = 7000\,km$ $\quad e = 0.0015$ $\quad \omega = 0^\circ$ $\quad i =$ $\quad u = \omega + M$ $(\delta\lambda = 0)$

WkL2

along track separation $2a\delta e \rightarrow ?$

Chief $(e_x, e_y) = (e\cos\omega, e\sin\omega)$

$= (0.0015\cos 0^\circ, 0.0015\sin 0^\circ) = (0.0015, 0)$

Deputy $(0.0015\cos 90^\circ, 0.0015\sin 90^\circ) = (0, 0.0015)$

$\delta\vec{e} = (-0.0015, 0.0015)$ $\quad |\delta\vec{e}| = 0.0015\sqrt{2} = 0.0021213$

along track separation $2a|\delta\vec{e}| = 2 \times 7000 \times 0.002143$

$= 29.6982\ km$

Q30) $\delta a = 0$ $a\delta e = 400m$ $a\delta i = 600m$ $\varphi - \vartheta = 40^\circ$

$d_{min} = ?$

$a\delta e = \vec{E}$ $a\delta i = \vec{I}$

$$\delta r_{min} = \frac{\sqrt{2}\,|\vec{E}\cdot\vec{I}|}{\sqrt{E^2 + I^2 + |\vec{E}+\vec{I}|\,|\vec{E}-\vec{I}|}}$$

$|\vec{E}\cdot\vec{I}| = E \cdot I \cdot \cos(\varphi - \vartheta) = 400 \times 600 \times \cos 40 = 183850.6663\ m^2$

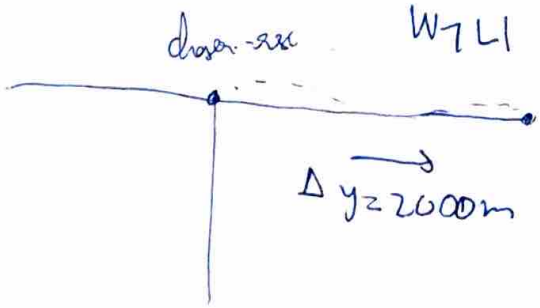
$|\vec{E}+\vec{I}| = \sqrt{(400)^2 + (600)^2 + 2(400)(600)\cos(49^\circ)} = 942.18\ m$

$|\vec{E}-\vec{I}| = 390.25\ m$

$$d_{min} = \frac{\sqrt{2} \cdot 183850.666}{\sqrt{400^2 + 600^2 + (942.18)(390.25)}}$$

$$= \frac{260064.1106}{945.166} = 275.09\ m$$

Q31) $a = 6778\,km$



$$\Delta y = -3\pi \Delta v_y \quad / \; n$$



$$\sqrt{\frac{\mu}{a^3}} = \sqrt{\frac{398600}{(6778)^3}} = 0.0011313\,9$$

2 tangential impulses
1 orbit apart



$$2\,km = -3\pi \cdot \Delta v_y \quad / \; 0.0011313\,9$$

$$\Delta v_y = \frac{2 \times 0.0011313\,9}{-3\pi} = -0.000240089\,97\ km/s$$

$$= -0.240089\ m/s$$

Q32) $a = 6678\,km$

$$\delta r_0 = [0, -3000, 0]^T\ m$$

$$\delta v_0^- = 0$$

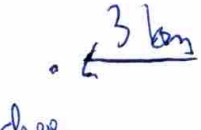
HCW STM $\rightarrow \Delta v$?

arriving in 1 hr

$$v = \sqrt{\frac{\mu}{r}} = 7.7258\ km/s$$

$$P = 2\pi \sqrt{\frac{a^3}{\mu}} = 5431.01\ s$$

$$n = \sqrt{\frac{\mu}{a^3}} = 0.0011568\ rad/s$$



$$dr_f = [0, 0, 0]^T$$

$$t_0 = 0$$

$$t_f = 3600\,s$$

WSL2

using hf tok $\rightarrow$ STMΦ

$$\Delta V \text{ at } t_0 = \Delta v_0 = -\left[\Phi_{rv}(t_f)\right]^{-1}\left[\Phi_{rr}(t_f)\right]\delta r_0$$

null velocity on arrival $= [-0.47099, -0.12687, 0]^T\ km/s$

$$\Delta v_f = \Phi_{vv}\,\delta v_0^+ = [0.47099, -0.12687, 0]^T\ km/s$$

$$\Delta v_{total} = 0.9756\ m/s$$

Nitya

Q3) $a = 7000$ km

$e = 0.2$

Chaser → $x_0 = 200$ m, $y_0 = 0$

$\dot{x}_0 = 0$

$$(2+e)x_0 + \frac{1+e}{\dot{\theta}_p}\dot{y}_0 = 0,$$

$$n = \sqrt{\frac{398600\times10^{9}}{(7000\times10^{3})^3}} = 0.001078 \text{ rad/s}$$

$$\dot{\theta}_p = n\sqrt{\frac{1+e}{(1-e)^3}} = n\sqrt{\frac{1+0.2}{(1-0.2)^3}}$$

$$= 1.530931 \cdot n$$

$$= 0.0016503 \text{ rad/s}$$

$$\dot{y}_0 = -\frac{(2+e)\dot{\theta}_p}{1+e}x_0 = -\frac{2+0.2}{1+0.2} \times 0.0016503 \times 200$$

$$= -0.605 \text{ m/s}$$